

SDLC Overview: Software Development Life Cycle

1. Requirements Gathering

- Understand what the stakeholders need from the system.
- Involves meetings, surveys, and documentation of functional and non-functional requirements.
- Critical because it sets the direction for the entire project.
- Forms the foundation for the design phase.

2. Design

- Create a plan or blueprint for how the system will be built.
- Includes system architecture, user interface designs, database design, and technology selection.
- Ensures all stakeholders understand how the system will look and behave.
- Guides developers during implementation.

3. Implementation (Coding)

- Developers write code based on the design documents.
- Involves building modules, integrating components, and writing algorithms.
- This is where the software takes shape.
- Outputs working software for testing.

4. Testing

- Verifies that the software functions as expected.
- Includes unit testing, integration testing, system testing, and user acceptance testing.
- Ensures bugs and errors are identified and fixed before release.
- Helps maintain software quality and reliability.

5. Deployment

- The final software is released to the users or production environment.
- Includes installation, configuration, and training if needed.
- Marks the beginning of real-world use and feedback.
- May lead to further improvements or iterations.